Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
Rectified gain @ k=96 (kcal/mol)
Predicted most stabilizing per domain
0
1
10
6.50
6.49
13.25
16.65
3.49
3.74
9.00
10.17
3.69
3.81
9.51
11.17
1.99
2.97
6.22
8.19
2.70
2.70
8.46
9.18
2.89
3.26
8.79
9.16
2.23
3.15
2.67
3.55
2.81
4.37
3.55
3.30
0.48
0.25
0.90
0.43
2.60
1.66
6.74
5.45
2.05
2.22
6.36
6.54
12.97
17.18
26.88
38.97
12.10
17.03
28.09
38.78
11.54
16.31
27.70
40.45
10.02
17.91
25.29
42.78
11.24
15.17
29.63
39.96
11.46
14.82
30.72
39.00
20.73
30.67
13.91
30.70
23.24
32.18
16.21
24.36
21.85
27.85
21.30
25.00
19.48
10.91
33.13
32.42
11.97
17.06
28.00
38.97
Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
Δ hydrophobicity ≤ 0 & relative ASA ≤ 0.5
Δ hydrophobicity > 0 & relative ASA ≤ 0.5
Δ hydrophobicity ≤ 0 & relative ASA > 0.5
Δ hydrophobicity > 0 & relative ASA > 0.5

Top 96 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,295 single mutants); gain = sum of max(ddG, 0) over the picks predicted stabilizing (> 0); boxes: interquartile range with median line; dots: domains; annotations: mean over domains; dashed lines: oracle mean.